

In 2014 a study was conducted by a PhD candidate at Georgia Southern University examining the correlation between swimming ability, gender, and race. Data from that study included the following information in a two-way frequency table. Use this information to answer questions 1-6.

Ability	Male	Female	Total
Cannot Swim	16.07%	18.50%	34.57%
Can Swim a Little	19.14%	17.11%	36.25%
Can Swim Efficiently	15.37%	13.81%	29.18%
Total	50.58%	49.42%	100.00%

1. Interpret the joint relative frequency females who can swim a little.
2. Interpret the value of 34.57% in context of the data in the table.
3. Interpret the value of 13.81% in context of the data in the table.
4. Interpret the value of $\frac{36.25\%}{100.00\%}$ in context of the data in the table.
5. Interpret the value of $\frac{19.14\%}{50.58\%}$ in context of the data in the table.
6. What percentage of females can swim efficiently based on the data in the table?
7. What percentage of those who can swim efficiently are males?
8. Can you make any associations between gender and swimming ability based on the data? Explain.

In the 2014 Georgia Southern University study the correlation between swimming ability and classification in school was also studied. Data from that study included the following information:

- 36.15% of all students were Sophomores.
- 24.34% of all students were Seniors.
- 36.27% of all students can swim a little.
- Of all students, 29.18% swim efficiently.
- 41.02% of seniors can swim a little.
- 7.91% of students that cannot swim are juniors.
- Of the students that cannot swim, 19.54% are sophomores.
- Of the seniors, 23.83% cannot swim.
- 23.75% of students that swim efficiently are juniors.
- Of those students that can swim efficiently, 19.72% are freshmen.
- 1.92% of students were freshmen that can swim a little.

Use this information to answer questions 9-10.

9. Based on the information above create a two-way relative frequency table. Round to the nearest hundredth.

Ability	Freshman	Sophomore	Junior	Senior	Total
Cannot Swim					
Can Swim a Little					
Can Swim Efficiently					
Total					

10. Does there appear to be an association between the abilities to swim and the class of students in school?